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THE DEMAND FOR MONITORING TECHNOLOGIES: THE CASE OF TRUCKING*

THOMAS N. HUBBARD

This paper examines the demand for on-board computers in trucking, distinguishing between their incentive- and resource-allocation-improving capabilities. I find that monitoring's incentive benefits are high when perquisite-taking is attractive to drivers, driver effort is important, and verifying drivers' actions to insurers is valuable. These results are consistent with agency theory and suggest that networking applications will raise the productivity and pay of difficult-to-evaluate workers. I also find that monitoring's benefits are disproportionately resource-allocation-related when managerial decisions are least constrained. This suggests that networking applications' monitoring capabilities raise the returns to delegation when resource allocation decisions are routine and lower them when they are not.

I. INTRODUCTION

The diffusion of information technology (IT) across firms during the past 50 years is a economic event of great significance. Although many of its microeconomic implications are not completely understood, it is clear that IT has changed how production is organized. For example, as a substitute for some skills and a complement to others, IT has redefined managers' and workers' jobs in many sectors of the economy [Autor, Katz, and Krueger 1998; Bresnahan, Brynjolfsson, and Hitt 1999].

Recent advances in networking technologies have the potential to initiate even greater organizational changes. While earlier generations of applications mainly exploit IT's information processing capabilities, networking applications exploit its communication capabilities to a large degree. The premise that these applications will profoundly affect organizations follows from economic theories of organizations. Costs of observing, verifying, or communicating information are at the core of virtually all of these theories. These costs determine compensation schemes, job definitions, and other organizational variables. Technologies that lower these costs therefore should directly change how production is organized.

One important class of networking applications enables

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managers to monitor production processes.[1] The demand for monitoring comes from two sources. First, monitoring can improve incentives. It can help firms align workers' pay with their productivity, for example. Second, monitoring can improve resource allocation decisions ("coordination"). Economic theory suggests that the organizational implications of monitoring depend on whether it improves incentives or coordination. If monitoring improves incentives, it tends to increase the returns to delegation: adopting firms may give better-monitored workers more responsibilities. If it improves coordination, and makes managers better informed relative to workers, it can decrease the returns to delegation. It follows that understanding the demand for monitoring is necessary for understanding some of the long-run organizational implications of the diffusion of networking technologies.[2]

This paper investigates demand for on-board computers (OBCs) in the trucking industry. There are two classes of OBCs: trip recorders and electronic vehicle management systems (EVMS). Trip recorders electronically record speed, idle time, and other variables related to how drivers operate trucks. Dispatchers download this information once trucks return to their base. EVMS have all trip recorders' capabilities, but can also track trucks' geographic location, send information to dispatchers in real time, and give dispatchers a means of contacting distant drivers. Trip recorders help dispatchers provide drivers better incentives. They do not improve coordination because they do not supply information in a timely enough fashion. The marginal benefit of EVMS over trip recorders is that EVMS enhance dispatchers' ability to make and implement resource allocation decisions.

Using truck-level data from 1992, I estimate the determinants of demand for trip recorders and EVMS. I use these estimates to test a series of propositions about the demand for monitoring. Applying agency theory to the context of trucking, monitoring's incentive benefits should be high when perquisite-

1. FedEx's package tracking and SAP's supply chain management applications are well-known examples. "Groupware" applications such as Lotus Notes also enable monitoring.

2. Gurbaxani and Whang [1991] and Jensen and Meckling [1992] make similar observations. Like this paper, these authors draw from both agency-based [Alchian and Demsetz 1972; Jensen and Meckling 1976; Holmstrom 1979, 1982; Grossman and Hart 1983; Holmstrom and Milgrom 1991, 1994; Baker 1992] and nonagency theories of the firm [Marschak and Radner 1972; Aoki 1986; Geanakoplos and Milgrom 1991; Bolton and Dewatripont 1994].

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taking is attractive to drivers, when on-time arrivals are important, and when verifying drivers' actions to third parties is valuable. Furthermore, contrasts between agency theory and decision theory imply that OBCs' benefits should be disproportionately incentive-related when verification is particularly important, and coordination-related when dispatchers' resource allocation decisions are least constrained. Dispatchers are least constrained when spot arrangements mediate trade between carriers and shippers.

I find evidence consistent with each of these propositions. Trip recorders are most valuable when trucks have infrequent scheduled stops, haul goods for which inventories tend to be low, or haul dangerous or hazardous cargo. Trip recorders' value relative to EVMS is higher when trucks haul dangerous or hazardous cargo than when they haul other goods. EVMS' value relative to trip recorders is higher when spot arrangements mediate trade than when longer-term arrangements do.

These results have several implications with respect to the organizational impact of networking applications. Monitoring technologies improve both incentives and coordination. Their incentive benefits are highest exactly where agency theory predicts: for workers whose performance is otherwise difficult to evaluate and where shirking is costly. An implication of agency theory is that adoption should raise these workers' productivity. If labor markets are competitive, their pay should increase as well. These results also suggest systematic differences in how the diffusion of monitoring applications will change managers' and workers' job definitions. Monitoring technologies will lead to more delegation when resource allocation decisions are routine and less delegation when they are highly adaptive. Future research will test these organizational implications.

The organization of the rest of the paper follows. In Section II, I characterize incentive and coordination problems in trucking and describe how OBCs can mitigate them. I then develop a series of testable propositions. In Section III, I describe the empirical framework. Here, I propose empirical tests of the propositions and of alternative interpretations of the estimates. In Section IV, I describe the data and report simple statistics. Section V reports the estimates of the empirical model and discusses the results of the tests. Section VI concludes.

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II. Incentives, Coordination, and On-Board Computers in Trucking

A. Incentives

Relationships between carriers and drivers are principal-agent relationships. Drivers maximize utility, which is a decreasing function of effort and an increasing function of income and nonpecuniary benefits. Carriers maximize profits. Drivers affect when they arrive at destinations and the cost of the haul by how they operate trucks. Costs are increasing and convex in speed; thus, for a given haul they increase with both average speed and its variance. Drivers' objectives differ from carriers' because arriving on time requires effort, the cost of which they bear privately. Also, driving consistently means that they forgo the opportunity to work at their own pace. Drivers may prefer to achieve on-time arrivals by speeding and then taking longer breaks. Perquisite-taking of this form is most attractive when drivers have few scheduled stops. They can take longer breaks when they stop infrequently because they have more room to make up time. First-best decisions are those that equate marginal benefits and costs, taking into consideration both drivers' and carriers' objectives. Agency costs arise because some of drivers' actions are not contractible.

Firms address incentive conflicts using two classes of incentive instruments. One is performance incentives. Carriers use output measures and arrival times to evaluate drivers.[3] These variables are observable at very low cost. Carriers can easily calculate the shortest distance between drivers' origin and destination; aggregating this across hauls provides an output measure. They can learn arrival times by calling drivers' destinations and asking when deliveries took place. When drivers fall behind schedule, they learn these even more cheaply because customers call in. Poor-performing drivers are paid less, receive less-desirable job assignments, and in extreme cases are fired.

Although performance incentives reduce agency costs, they do not eliminate them. One reason is that basing incentives on arrival times forces drivers to bear risk associated with factors

3. Other variables are potentially informative, but are less useful because they are noisy. Trucks' engines are affected by how drivers operate them, but it is hard to distinguish whether depreciation or breakdowns are due to how trucks are operated or maintained. Drivers who never speed get into fewer accidents, but accidents are rare and can be due to other drivers' actions.

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outside of their control such as traffic and equipment malfunctions. The other is that they do not address the issue of driving variability within a haul. Drivers do not bear all of the costs associated with how they drive. Assuming that performance incentives are set optimally, drivers drive more variably, bear more risk, and have weaker incentives to arrive on time than they would under a "first-best" contract.

The other incentive instrument is asset ownership. Some drivers own their trucks. Drivers who own their trucks internalize most of the costs associated with how they drive, particularly because they pay for their own fuel as well. This institutional response is not prevalent. During 1992 only about 10 percent of tractor-trailers were owned by their drivers. There is apparently a trade-off involved in using asset ownership as an incentive instrument. Several researchers have proposed that this trade-off arises because asset ownership increases drivers' incentive to engage in rent-seeking behavior.[4] According to these theories and anecdotes from the industry, asset ownership improves drivers' incentives within hauls but worsens them between hauls.

Monitoring how drivers reach their destination can reduce agency costs within carrier-driver relationships in two ways. First, it can help carriers motivate drivers to work at the pace carriers prefer. This is particularly important when drivers do not own their trucks. Second, monitoring can reduce agency costs associated with rewarding drivers on the basis of arrival times. Better information about why drivers arrive late allows carriers to provide drivers stronger performance incentives without shifting as much risk to them.

Incentive conflicts also exist between carriers and other parties. For example, agency problems exist between shippers and carriers. This paper does not focus on how monitoring can reduce agency costs within these relationships. Interviews and a review of the trade press provide no evidence that firms used information collected by OBCs to improve carriers' incentives during the sample period. For example, contracts were not contingent on this information. For the rest of the paper the analysis will proceed assuming that carriers' and shippers' interests coincide. These outside sources do suggest that OBCs were used to improve contracts between carriers and another class of third parties: insurers. Agency costs arise because insurance

4. Nickerson and Silverman [1998] and Baker and Hubbard [1999].

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contracts are incomplete. It is often difficult to determine who is at fault when accidents occur: truck drivers or other drivers. Information that provides verification in such circumstances can improve insurance contracts. This is particularly valuable when accidents are most costly—such as when trucks haul dangerous or hazardous cargo.

B. Coordination

Coordination in this context is the process through which trucks and drivers are matched to hauls. Coordination issues are purely about resource allocation. Dispatchers play an important coordinating role within firms. They arrange schedules and job assignments in a changing environment in which they are imperfectly informed. New orders come in throughout the day. Unanticipated delays from traffic and crowded loading docks can disrupt schedules. Dispatchers respond by adjusting schedules and rearranging job assignments.

Learning trucks' status and communicating with drivers have traditionally been straightforward for local hauls, but not long-distance hauls. Dispatchers communicate with local drivers using two-way radios. These have a range of about 25 miles. Radio-based systems provide a simple way dispatchers can obtain information about trucks' status and redirect drivers, but have the usual drawbacks associated with systems where information exchanges are oral. Dispatchers must receive information at the same time drivers communicate it (and vice versa), and dispatchers receive information in a form that cannot be directly processed electronically. For longer hauls, dispatchers and drivers have traditionally used a "check-and-call" system whereby drivers call in when they make pickups or deliveries, or every two to three hours when stops are infrequent.[5] This system has two additional drawbacks relative to radio-based systems. First, dispatchers cannot initiate communication with drivers. This means they can neither obtain updated information about trucks' status nor redirect drivers until drivers call in. Second, drivers must pull over and find a pay phone when they call in. This slows production.

Coordination issues also arise between shippers and carriers. Logistics managers base resource allocation decisions on forecasts

5. Cellular phones were infrequently used by long-haul drivers during this time because of expensive "roaming charges."

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of when trucks will arrive. These decisions schedule loading docks and cargo handlers. Resources are underutilized when trucks arrive unexpectedly early or late. Logistics managers can obtain forecasts of arrival times by calling carriers, but these forecasts are only as good as carriers' information. Shippers and carriers can coordinate better when carriers have good information about trucks' location.[6]

Few economic theories of organizations revolve around coordination rather than incentive issues. Analysis of the value of information in such contexts follows from decision theory.[7] When incentives are not an issue, the value of informational improvements depends on (i) how much information improves individuals' beliefs about the state of the world, and (ii) holding constant changes in beliefs, how valuable improved decisions are at the margin. Note that agency theory suggests similar statements about information's incentive benefits. For the most part, theories do not generate predictions regarding where monitoring should be primarily incentive- or coordination-improving. There are two exceptions to this, however.

One is that information's timelines is generally less important for incentive than coordination improvements. Variables need not be observed in real time to improve contracts. In contrast, timeliness can be critical for improving resource allocation decisions, particularly when decisions are made in rapidly changing environments. The other is that verifiability is sometimes important for incentive improvements, but rarely so for coordination improvements. Information sometimes must be in a form verifiable to third parties (such as courts) to improve contracts. It generally need not be to improve resource allocation decisions.

C. On-Board Computers

On-board computers appeared on the market in the mid-late 1980s. There are two classes: trip recorders and electronic vehicle management systems (EVMS). Each of these are aftermarket

6. This paper does not consider how information technology improves matching between shippers and carriers in the market. While it may do so in the long run, there is no evidence that the information OBCs supplied directly facilitated better matches between shippers and carriers during the time I study. These devices provided carriers information about their own capacity, but this information was not passed on to the market at large (unlike, for example, on-line air reservation services).

7. See Raiffa and Schlaifer [1961], Raiffa [1968], and DeGroot [1970] for expositions of basic decision theory.

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equipment that can be installed or removed from trucks in two hours or less, usually by a mechanic or equipment dealer. Low-end trip-recorders cost about $500 in 1992. EVMS hardware cost $3000–$4500 to buy or about $150/month for a long-term lease.

Trip recorders chronicle trucks' operation. They record variables such as when drivers turn trucks on and off, how long trucks idle, and their speed. This information can be used to improve incentives. Carriers can obtain better measurements of how drivers drive and whether they are delayed by factors outside of their control. Insurers and courts can verify how drivers were driving at the time accidents took place. Trip recorders also record data that track the performance of trucks' engines—particularly engines with electronic controls. This helps mechanics diagnose engine problems. Finally, drivers can electronically record fuel purchases and when they cross state lines on trip recorders. This can lower the cost of filing state taxes or fulfilling other regulatory requirements. Dispatchers receive the data trip recorders collect once trucks return to their base.

EVMS generally have all trip recorders' capabilities plus several important additional features. First, they record trucks' geographic location, often through links to global positioning systems. Second, they can transmit the information they collect to dispatchers in close to real time. Dispatchers can have up-to-date information on individual trucks' location. Third, they provide a means whereby dispatchers can initiate communication with drivers while they are on the road. Dispatchers can send short text messages that instruct drivers or ask them to call in.

There is an economically important distinction between these classes of devices. Trip recorders are useful for improving incentives but not coordination. They are not useful for improving coordination because they do not provide information in a timely enough fashion: dispatchers make important resource allocation decisions in real time during the day.[8] In contrast, EVMS are useful for improving both incentives and coordination, because they enable real-time monitoring and a new way of communicating with drivers. I exploit this distinction when developing empirical tests.

8. The information these devices collect could in principle be used to help in benchmarking. For example, knowing how long particular hauls tend to take would improve scheduling decisions that are not made in real time. However, these devices are rarely used for such purposes. Carriers have less expensive ways of benchmarking routes.

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Factors that raise monitoring's incentive benefits increase the value of using trip recorders. This is the basis of the following testable propositions.

First, trip recorders should be more valuable when trucks stop infrequently. Therefore, they should be more valuable for trucks that operate far from home than those that operate close to home. Those that operate close to home have more frequent scheduled stops and return to their base at the end of the day. Similarly, holding distance constant, trip recorders should be more valuable on trucks used for less-than-truckload (LTL) hauls than for truckload (TL) hauls. Trucks used for local LTL hauls (such as UPS trucks) stop many times during the day, sometimes stopping at terminals to load or unload goods. Local TL trucks haul goods from point to point without stopping at terminals during the day. Long distance LTL trucks have regular routes and stop at terminals at the end of each day; long distance TL trucks have irregular routes and do not stop at terminals at the end of each day.

Second, they should be more valuable when late arrivals are costly. They should therefore be more valuable for trucks that deliver to loading docks than those that do not. Loading docks and handlers of goods tend to be scheduled, particularly for refrigerated goods. Late arrivals mean that these resources are underutilized. In contrast, late arrivals are less disruptive for goods that can be dumped or poured. Trip recorders should also be more valuable when trucks haul goods for which inventories tend to be low. As suggested by the business literature on just-in-time inventory systems, lateness is more likely to disrupt production schedules in such circumstances.

Third, they should be more valuable when accidents are particularly costly. They should therefore be more valuable on trucks that haul dangerous or hazardous cargo such as chemicals and petroleum than on trucks that haul other goods.

In contrast, factors that raise (lower) monitoring's coordination benefits relative to its incentive benefits make EVMS more (less) valuable relative to trip recorders. Following the discussion of timeliness and verification above, this leads to two additional propositions.

First, EVMS' relative benefit should be higher when dispatchers face few constraints in how they respond to changes in short-run market conditions. There are two reasons for this. One is that the constraints themselves lower OBCs' coordination

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benefits. The other is that factors that lower the cost of constraining dispatchers also lower OBCs' coordination benefits. For example, one would expect there to be more constraints and fewer EVMS in circumstances where a truck's highest-value use is to serve a particular shipper consistently.[9] EVMS' relative benefit therefore should be higher within for-hire than private fleets. Private fleets exist primarily to serve internal customers. Dispatchers respond to external demand only in certain circumstances, such as when internal customers do not exist for long return trips ("backhauls"). Among for-hire trucks, EVMS' relative benefit should be higher when trucks haul under common carriage agreements than under contract carriage agreements. Many motor carrier contracts constrain dispatchers' resource allocation decisions by dedicating part of carriers' fleet to serve individual shippers. Dispatchers often cannot use certain trucks and drivers to serve other shippers without permission. In contrast, dispatchers face few explicit constraints within common carriage. Dispatchers are generally free to allocate resources to meet demand as they wish when common carriage arrangements mediate trade.

Second, the relative benefits of EVMS should be low when monitoring is primarily valuable for verification purposes. All else equal, they therefore should be lower when trucks haul goods in tank trucks than otherwise, because tank trucks haul petroleum and chemicals.

Testing these propositions requires an empirical framework through which one can detect cross-haul differences in trip recorders' value, and in EVMS' value relative to trip recorders. I develop such a framework in the following section.

III. Empirical Framework

A. Revealed Preferences and the Allocation of Trucks

Suppose that carriers and shippers make long-, medium-, and short-run decisions, where the magnitude of adjustment costs distinguishes the categories. They make these decisions toward maximizing individual profits. Long-run decisions include firms' hierarchies, investments in immovable physical capital such as terminals, investments in backoffice IT, and the like. Firms make

9. Constraints would serve to mitigate carriers' threat to use trucks in lower-valued uses. See Hubbard [1999] for more on specificity and contractual form in trucking.

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medium- and short-run decisions taking them as given. Medium-run decisions determine the size and composition of carriers' fleets. These decisions establish whether trucks have OBCs and whether drivers own them. Medium-run decisions also determine whether shippers and carriers mediate trade using long-term arrangements. Carriers and shippers base medium-run decisions on market forecasts, anticipating that short-run production-related decisions will be optimal, conditional on the medium- and long-run decisions.

Suppose that there are three kinds of trucks: trucks without OBCs, with trip recorders, and with EVMS. Assuming that carriers are price-takers in the market for each truck type and shippers and carriers do not make systematic forecasting errors, the allocation of trucks across hauls will be efficient from a medium-run perspective.[10] One can then employ a revealed-preference-based empirical framework to test the propositions above.

A revealed-preference-based framework carries with it the assumption that shippers and carriers are able to assess OBCs' value on particular hauls accurately. This is not always a safe assumption for new technologies. The computer industry in particular contains many examples in which new technologies' most valuable uses were not immediately apparent.[11] However, uncertainty was relatively low for OBCs. The most important applications to date—monitoring drivers toward improving incentives, scheduling, and maintenance—were recognized and widely reported in the trucking trade press very soon after the devices appeared on the market. By 1992, only five years after they first came on the market, OBCs were recognized as an important technology, and their monitoring capabilities were well-understood. Although some firms may have remained uncertain about their value, uncertainty did not affect adoption patterns nearly as much as it does in the computer industry in general.

10. If competitive short-term rental markets existed for each type of truck, and carriers rented their marginal trucks of each type, the allocation of trucks would be efficient in the short run as well. However, while trucks with OBCs preinstalled were available for lease on a long-term basis, they were not available as short-term rentals during 1992.

11. For example, there was considerable uncertainty about what were the most valuable uses of the internet—even in the near term—during the period it was first widely diffusing.

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B. Empirical Specification and Alternative Hypotheses

Let y^*_{1i}, y^*_{2i}, and y^*_{3i} be the net benefit of using a truck without an OBC, with a trip recorder, and with EVMS, respectively, for haul i. The net benefits are calculated across all interested parties: carriers, shippers, and drivers. Let

$$y^*_{si} = X_i\beta_s + \epsilon_{si},$$

where $s = \{1,2,3\}$. I assume that $\epsilon_i = \{\epsilon_{1i},\epsilon_{2i},\epsilon_{3i}\} \sim N(0,\Sigma)$. Normalizing $\beta_0 = 0$ and $\Sigma_{11} = \Sigma_{22} = \Sigma_{33} = 1$, one can estimate β_2, β_3, and $\omega_1 = \text{cov}(\epsilon_{2i} - \epsilon_{1i},\epsilon_{3i} - \epsilon_{1i}) = 1 + \Sigma_{23} - \Sigma_{13} - \Sigma_{12}$ using a multinomial probit. The advantage of using a multinomial probit rather than a nested logit in this case is that it does not constrain any of the covariance terms Σ_{12}, Σ_{13}, and Σ_{23} to be positive, regardless of how other elements of Σ are normalized.

In the basic specification, X_i includes carrier and haul characteristics. One important variable that is included in X_i but is endogenous in the medium run is whether owner-operators are used. The fact that asset ownership exists as an incentive substitute complicates matters, because it can mean that OBC use patterns do not correspond to differences in monitoring's incentive benefits. For example, suppose that asset ownership decreases agency costs disproportionately relative to using trip recorders as haul length increases. Then one may not observe that trip recorders are used more for longer hauls even if their incentive benefits increase. A maintained assumption throughout the rest of this paper is that incentive substitutes such as asset ownership decrease agency costs proportionately relative to using OBCs. Under this assumption, if a factor increases monitoring's incentive benefits, trip recorder use will increase with the factor.[12] One can think of this as a condition on the technical rate of substitution that is analogous to homotheticity in neoclassical theory. Figure I shows evidence consistent with this assumption: the fraction of trucks driven by owner-operators and the fraction of trucks equipped with OBCs increase roughly proportionately with how far from home they operate.

Assuming that $E(X_i\epsilon_{si}) = 0$ and trucks are homogeneous except in how they are equipped, β_2 and β_3 indicate how the net benefits of using trip recorders and EVMS, respectively, vary with X_i. I report estimates of β_2 and of $(\beta_3 - \beta_2)$ in the results tables.

12. Furthermore, this holds regardless of whether one includes owner-operator as an explanatory variable.

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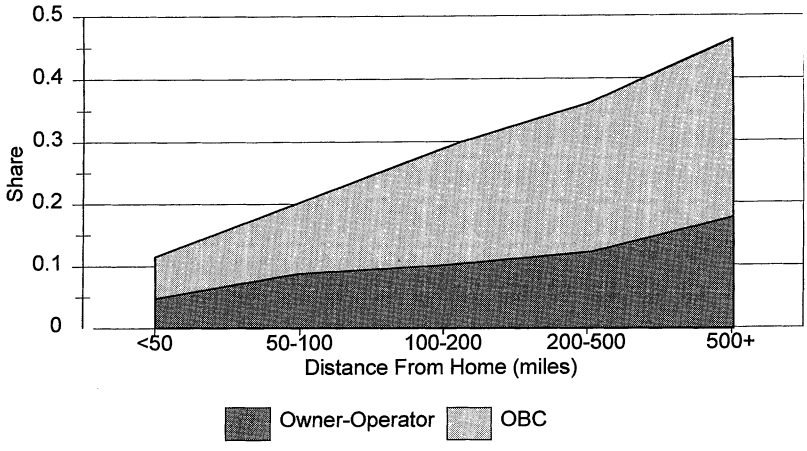


FIGURE I
Owner-Operator and OBC Shares

Tests of propositions regarding monitoring's incentive benefits are based on the null H_0:$\beta_2 = 0$. Positive estimates of β_2 indicate situations where the value of trip recorders increases with X_i; these factors increase monitoring's incentive benefits. Tests of propositions regarding the nature of monitoring's benefits are based on the null H_0: $\beta_2 = \beta_3$. Positive estimates of $(\beta_3 - \beta_2)$ indicate situations where OBCs' coordination benefits increase with X_i relative to their incentive benefits.

There are two important alternative interpretations of the coefficients that arise when the assumptions about truck homogeneity and the orthogonality of omitted variables fail to hold.

One alternative interpretation is that β_2 and β_3 reflect the value of using newer, more reliable trucks rather than the value of using OBC-equipped trucks. I examine this by running additional specifications that control for the vintage of the truck. If β_2 and β_3 reflect the value of using new trucks, the estimates should move toward zero when controlling for vintage. If the estimates do not change, this supports the interpretation that they reflect differences in the value of monitoring.

Another alternative interpretation is that β_2 and β_3 reflect differences in shippers' value of carriers' nonmonitoring-related technological capabilities. These capabilities are not directly observed in the data. For example, suppose that OBCs and electronic data interchange (EDI) capabilities are neither comple-

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ments nor substitutes, but adoption is correlated across carriers.[13] One reason for the latter might be that some carriers are technologically sophisticated, and this lowers their cost of developing and implementing applications. Then patterns in OBC use may reflect differences in the value of EDI—differences that are assumed to be unrelated to the value of monitoring. Although including controls for fleet size probably accounts for much of the variance in carriers' other technological capabilities, it does not account for differences among similarly sized carriers. Omitted carrier capabilities may bias the estimates so that they do not reflect differences in the value of monitoring.

I examine this interpretation by investigating whether the patterns from the full sample hold when restricting the analysis to trucks in very small fleets. Very small fleets uniformly had few nonmonitoring-related technological capabilities as of 1992. These carriers did not use EDI to send shippers electronic invoices, for example.[14] Patterns of OBC use within this subsample likely do not reflect unrelated technological capabilities. If the results of the hypothesis tests on this subsample are similar to those on the full sample, this suggests that the patterns in the full sample reflect differences in the value of OBCs, not just in the value of unrelated technological capabilities.[15]

IV. Data

The data are from the Census Bureau's 1992 Truck Inventory and Use Survey (TIUS). This is a mail survey sent to about 150,000 truck owners; these are selected at random from vehicle registrations. Truck owners are required by law to respond. This paper uses only observations of truck-tractors, the front part of tractor-trailer combinations (N = 39,850).

The TIUS collects data on the attributes of individual truck-tractors: their characteristics as they were built and any aftermarket equipment. This includes whether trucks have trip recorders or EVMS installed. In some cases, respondents record that

13. One EDI capability unrelated to OBCs is the ability to exchange billing information electronically.

14. EDI diffusion was low even among the largest carriers during this time. See Crum, Johnson, and Allen [1998].

15. Note that this paper is not concerned with distinguishing between the value of OBCs and that of capabilities that are strictly complementary, such as shipment-tracing software. OBC adoption is observationally equivalent to OBC-plus-complementary-software adoption if they are always used jointly.

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both are installed. I assume that these respondents are reporting their OBCs' capabilities, and treat circumstances where respondents indicate "both" the same as where they indicate "EVMS only." The TIUS also collects variables that reflect how trucks are used. These variables report how far from home trucks generally operate, what type of trailers are commonly attached to them, what class of products they generally haul, and whether they are part of private or for-hire fleets. When trucks are part of for-hire fleets, the data also report whether they are used for TL or LTL hauls and whether they are used for contract or common carriage. The TIUS also classifies trucks according to the size of the fleet of which they are part. But the data do not report detailed information about trucks' owners, and one cannot link trucks to carriers due to confidentiality restrictions.

To test hypotheses about how inventory levels affect the value of monitoring, I merge the TIUS with data from the 1992 Census of Wholesale Trade. The latter contains information on the sales/inventory ratios of wholesale firms, by product class. I interpret sales/inventory ratios at wholesale firms as proxies for sales/inventory ratios for the product in general.[16] I also incorporate data from the Census of Manufacturers that report the distribution of manufacturers' shipments: the shares shipped to wholesalers and retailers, by product class. These data pick up differences in where products are located within supply chains, and are included in specifications as controls. Details regarding these outside sources and the merging process are in Appendix 1; Appendix 2 contains these series.

Table I reports cross-tabulations. Overall, 7.5 percent of trucks in my sample have trip recorders installed, and 10.5 percent have EVMS installed.[17] Both devices are more common on trucks used for long hauls than short ones, especially EVMS. Among for-hire trucks, OBCs are more common on those used for TL than LTL hauls. OBCs are used more when trucks stop infrequently. Both classes of OBCs are more common on trucks attached to van-type trailers than on those attached to platforms or specialized trailers. Trip recorders are more common and EVMS less common on trucks attached to tank trucks than those

16. A better measure of cross-product differences in inventory management would contain information regarding manufacturers' and retailers' inventories. However, I have been unable to find data series that report this information by product class rather than manufacturer or retailer class.

17. All sample means, cross-tabulations, and estimation results use expansion factors provided by the Census as weights.

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TABLE I
TRIP RECORDER AND EVMS USE RATES

Category	Percent of sample	Variable	Use rate (percent)	
			Trip recorder	EVMS
	100.00	All Truck-tractors	7.51	10.54
Distance from home	26.55	<50 miles	3.94	2.33
	17.47	50–100 miles	6.92	4.35
	13.78	100–200 miles	11.28	7.67
	17.83	200–500 miles	10.82	13.54
	24.38	500+ miles	6.96	21.98
TL/LTL (for-hire only)	72.69	TL	6.82	17.18
	20.08	LTL	5.16	10.14
	7.23	Missing	5.46	7.95
Type of trailer	8.93	Tank truck	14.23	7.49
	11.40	Refrigerated van	13.46	20.88
	32.70	Enclosed nonrefrigerated van	7.20	12.83
	22.43	Platform	4.46	5.57
	19.36	"Specialized trailer" (all others)	4.64	5.60
Carriage type	50.06	Private carriage	8.83	5.84
	19.96	Contract carriage	7.44	13.35
	23.60	Common carriage	5.92	18.36
	2.93	Exempt carriage	4.13	5.60
	3.57	Missing	5.41	7.93
Truck owner	86.98	Carrier ("Company driver")	8.32	11.16
	10.85	Driver ("Owner-operator")	2.04	4.89
	2.17	Missing	6.82	8.39
Fleet size (trucks and trailers)	49.58	<25	3.04	4.11
	19.36	25–99	8.96	8.91
	14.88	100–499	15.30	14.34
	3.95	500–999	14.99	21.82
	6.21	1000–4999	14.29	27.44
	1.80	5000–10000	8.95	33.35
	4.23	>10000	5.68	21.84
Model year	52.31	pre-1987	5.46	2.63
	7.81	1987	8.75	4.93
	8.25	1988	9.55	9.22
	8.98	1989	10.04	13.38
	7.35	1990	11.12	16.93
	6.57	1991	10.03	26.32
	8.72	1992–1993	8.44	40.54

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attached to enclosed nonrefrigerated vans. Trip recorders are more prevalent and EVMS less prevalent on trucks within private than for-hire fleets, and within contract carriage than common carriage.[18] These facts suggest that OBCs' coordination benefits are low relative to its incentive benefits when accidents are particularly costly, and are high when dispatchers' decisions are least constrained. In general, the cross-tabulations are consistent with each of the propositions above. The results below test the propositions more formally.

From the bottom of the table, OBCs are used more in larger fleets and on more recent vintage trucks. These patterns probably are not entirely due to differences in the value of monitoring. This illustrates the importance of investigating the alternative hypotheses described in the previous section in which differences in OBC use reflect factors other than the value of monitoring.

V. Results

Table II contains estimates from the basic specification. Estimates of β_2 are on the left. Estimates of $(\beta_3 - \beta_2)$ are on the right. Most of the right-hand-side variables are either self-explanatory or defined above. Two that are not are "intrastate" and "private fuel." These serve as controls. "Intrastate" equals one if the truck is used primarily for intrastate hauls and zero otherwise. "Private fuel" equals one if the truck is refueled primarily at private fueling places (e.g., terminals) and zero if it is refueled at truck stops or service stations.

The estimates of β_2 support the proposition that monitoring's incentive benefits are highest for hauls where perquisite-taking is most attractive to drivers. The coefficients on the distance from home dummies are all positive and significant, and increase in magnitude up to 200 miles. Monitoring's incentive benefits are highest for trucks that operate far from home. The positive and significant coefficient on "TL" indicates that the incentive benefits are higher for truckload hauls than less-than-truckload hauls. Trip recorders are used more when trucks stop infrequently.

These estimates also provide evidence that monitoring's

18. "Exempt" carriage includes hauls that were traditionally exempt from Federal regulation. The data do not report the contractual form under which these hauls took places.

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TABLE II
MULTINOMIAL PROBIT RESULTS
Basic Specification (N = 39,850)

	Trip recorder		Difference ($\beta_3 - \beta_2$)	
	Estimate	Std. err.	Estimate	Std. err.
C	**−2.791**	**0.072**	**0.406**	**0.105**
50–100 miles	**0.142**	**0.035**	**0.121**	**0.055**
100–200 miles	**0.373**	**0.035**	0.043	0.055
200–500 miles	**0.392**	**0.034**	**0.263**	**0.052**
500+ miles	**0.320**	**0.038**	**0.553**	**0.056**
TL	**0.135**	**0.037**	**0.436**	**0.052**
Tank truck	**0.283**	**0.036**	**−0.458**	**0.057**
Refrigerated van	**0.179**	**0.035**	0.051	0.052
Platform	**−0.066**	**0.033**	**−0.103**	**0.048**
Specialized trailer	−0.058	0.033	0.010	0.048
Sales/inventory	**0.0151**	**0.0021**	−0.0040	0.0031
Private fleet	**0.413**	**0.037**	**−0.347**	**0.053**
Contract carriage	**0.132**	**0.032**	**−0.264**	**0.045**
Exempt carriage	0.129	0.081	**−0.299**	**0.113**
Intrastate	0.014	0.039	**−0.370**	**0.057**
Owner-operator	**−0.264**	**0.049**	**−0.236**	**0.064**
Private fuel	0.407	**0.025**	**−0.181**	**0.036**
Share wholesale	0.0015	0.0011	**−0.0038**	**0.0016**
Share retail	**−0.0031**	**0.0014**	0.0002	0.0020
Fleet size 25–99	**0.301**	**0.030**	**−0.135**	**0.044**
Fleet size 100–499	**0.599**	**0.030**	**−0.254**	**0.044**
Fleet size 500–999	**0.659**	**0.043**	**−0.152**	**0.064**
Fleet size 1000–4999	**0.685**	**0.039**	−0.018	0.057
Fleet size 5000–10000	**0.475**	**0.067**	**0.513**	**0.093**
Fleet size 10000+	**0.174**	**0.053**	**0.619**	**0.073**
Missing sales/inventory	**0.378**	**0.049**	**−0.263**	**0.072**
Missing share wholesale/retail	**−0.128**	**0.050**	−0.025	**0.072**
Omega1	**−0.878**	**0.082**		
−Log L		19432.90		

Difference denotes the difference in the parameter estimates on the EVMS and trip recorder branches. Boldface indicates rejection of H_0: $\beta = 0$ using a two-tailed t-test of size 0.05.

incentive benefits increase with the importance of on-time arrivals. The coefficient on the platform dummy is negative and significant using a t-test of size 0.05; that on the specialized trailer dummy is as well using a t-test of size 0.10. Trip recorders are less

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valuable for hauls that use these trailers than those that use enclosed, nonrefrigerated vans (the omitted category). Setting aside tank trucks, monitoring's incentive benefits are higher for hauls that deliver to loading docks than those that do not. The positive and significant coefficient on sales/inventory indicates that trip recorder use is highest when trucks haul goods for which inventories tend to be low. Monitoring's incentive benefits are thus high when deliveries are on tight schedules and when late arrivals are likely to cause stockouts.

Finally, the estimates indicate that trip recorders' value increases with the costliness of accidents. The tank truck coefficient is positive and significant. Monitoring's incentive benefits are high when third party verification is important.

The estimates of $(\beta_3 - \beta_2)$ provide evidence that OBCs' incentive benefits are high relative to their coordination benefits for hauls using tank trucks. The tank truck coefficient is negative and significant. OBCs' benefits are disproportionately incentive-related when monitoring is primarily important for third party verification.

They also indicate that OBCs' coordination benefits are high relative to their incentive benefits within for-hire fleets, particularly within common carriage. The coefficients on private fleet and contract carriage are negative and significant. OBCs' incentive benefits are high relative to their coordination benefits within private fleets and contract carriage, compared with the omitted category: common carriage. OBCs' benefits are disproportionately coordination-related when dispatchers face few constraints in how they respond to changes in the market.

Together, these results suggest a guide for the application of different classes of organization theory. They also suggest systematic differences in how one would expect new monitoring technologies to affect the benefits of delegation, and thus job design. Theories in which information improves incentives—such as agency theory—apply best when monitoring technologies primarily provide a means of verification and when resource allocation decisions are routine. Drawing from these theories, new monitoring technologies will tend to increase the returns to delegation in such circumstances. Adoption will encourage firms to give workers more responsibilities because it improves their ability to evaluate workers' decisions. In contrast, theories in which information improves coordination apply best when resource allocation

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decisions are highly adaptive. New monitoring technologies tend to decrease the returns to delegation in these environments. If the purpose of delegation is to capture informational advantages that workers have about their local environment, adoption will encourage firms to move decision-making authority from workers to managers.

Other estimates of $(\beta_3 - \beta_2)$ are not related to any hypothesis tests, but indicate how the nature of OBCs' benefits differs across hauls. Almost all of the coefficients on the distance dummies are positive and significant. The estimates tend to increase with distance. The coefficient on the TL dummy is positive and significant as well. Combined, these indicate that EVMS' value grows relative to trip recorders' when trucks operate farther from home and stop less frequently. OBCs' incentive and coordination benefits both increase with these variables. The coefficients on the fleet size dummies are negative and significant for small fleets and then turn positive and significant for large ones. One interpretation is that OBCs' benefits become disproportionately coordination-related as fleet size increases. Better information about supply may be more valuable when there are more ways of reoptimizing assignments as new information about demand comes in. Alternatively, the fleet size coefficients may indicate that EVMS are complementary to large fleets' other informational capabilities but trip recorders are not.

A. Alternative Hypotheses

Table I indicates that OBCs tend to be installed on newer trucks. I examine the hypothesis that the estimates in Table II reflect differences in the value of using new trucks rather than IT-equipped trucks by comparing them with estimates from a specification that controls for vehicle age. These are in Table III. The truck age coefficients indicate that EVMS are more common on more recent vintage trucks, but there is no relationship between trip recorder use and truck age. The only coefficients that change importantly are the coefficients on the distance dummies in the right panel, which become smaller. Controlling for trucks' vintage, EVMS do not become much more common relative to trip recorders as distance from home increases. EVMS' coordination benefits may not increase with distance as much as the results from Table II suggest. None of the results of the hypothesis tests described above change.

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TABLE III
MULTINOMIAL PROBIT RESULTS
Basic Specification (N = 39,850)

	Trip recorder		Difference ($\beta_3 - \beta_2$)	
	Estimate	Std. err.	Estimate	Std. err.
C	**−2.585**	**0.143**	**1.483**	**0.189**
50–100 miles	**0.135**	**0.035**	0.005	0.052
100–200 miles	**0.374**	**0.036**	−0.043	0.052
200–500 miles	**0.425**	**0.037**	**0.097**	**0.049**
500+ miles	**0.381**	**0.045**	**0.159**	**0.057**
TL	**0.219**	**0.046**	**0.330**	**0.055**
Tank truck	**0.253**	**0.040**	**−0.460**	**0.062**
Refrigerated van	**0.210**	**0.036**	0.050	0.044
Platform	**−0.070**	**0.033**	0.002	0.044
Specialized trailer	−0.069	0.033	**0.123**	**0.044**
Sales/inventory	**0.0154**	**0.0021**	−0.0035	0.0029
Private fleet	**0.415**	**0.037**	**−0.343**	**0.054**
Contract carriage	**0.112**	**0.034**	**−0.247**	**0.043**
Exempt carriage	0.099	0.081	**−0.269**	**0.105**
Intrastate	−0.032	0.043	**−0.269**	**0.058**
Owner-operator	**−0.308**	**0.050**	−0.033	0.060
Private fuel	**0.415**	**0.037**	**−0.171**	**0.033**
Share wholesale	0.0018	0.0011	**−0.0043**	**0.0014**
Share retail	**−0.0044**	**0.0015**	0.0032	0.0019
Fleet size 25–99	**0.291**	**0.030**	**−0.208**	**0.043**
Fleet size 100–499	**0.609**	**0.030**	**−0.338**	**0.050**
Fleet size 500–999	**0.691**	**0.044**	**−0.245**	**0.061**
Fleet size 1000–4999	**0.771**	**0.043**	**−0.158**	**0.048**
Fleet size 5000–10000	**0.608**	**0.077**	**0.340**	**0.083**
Fleet size 10000+	**0.267**	**0.066**	**0.363**	**0.076**
Truck age	−0.035	0.029	**−0.291**	**0.036**
Truck age**2	0.001	0.002	0.015	0.002
Missing sales/inventory	**0.361**	**0.050**	**−0.250**	**0.069**
Missing share wholesale/retail	**−0.130**	**0.051**	0.008	0.066
Omega1	**0.349**	**0.188**		
−Log L		18328.06		

Difference denotes the difference in the parameter estimates on the EVMS and trip recorder branches. Boldface indicates rejection of H_0: $\beta = 0$ using a two-tailed t-test of size 0.05.

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Table IV reports results when using only data from fleets with fewer than 25 trucks. Most of the estimates of β_2 are similar to above. Trip recorders are used more as distance from home increases up to 200 miles, then flattens out thereafter. The coefficients on sales/inventory and tank truck remain positive and significant, and are of about the same magnitude as in Table II. One difference from Table II is that neither the platform nor the specialized trailer coefficient is negative and significant. Although

TABLE IV
MULTINOMIAL PROBIT RESULTS
Fleet Size <25 (N = 21,041)

	Trip recorder		Difference ($\beta_3 - \beta_2$)	
Branch	Estimate	Std. err.	Estimate	Std. err.
C	**−2.554**	**0.126**	0.066	0.176
50–100 miles	−0.070	0.055	**0.417**	**0.083**
100–200 miles	**0.146**	**0.058**	**0.307**	**0.089**
200–500 miles	**0.167**	**0.059**	**0.593**	**0.086**
500+ miles	**0.223**	**0.065**	**0.623**	**0.093**
TL	−0.093	0.073	0.151	0.097
Tank truck	**0.207**	**0.069**	−0.172	0.099
Refrigerated van	**0.177**	**0.070**	−0.020	0.094
Platform	−0.020	0.053	−0.093	0.074
Specialized trailer	0.071	0.053	−0.102	0.075
Sales/inventory	**0.0114**	**0.0036**	**−0.0106**	**0.0050**
Private fleet	0.098	0.072	**−0.233**	**0.097**
Contract carriage	0.120	0.065	−0.065	0.084
Exempt carriage	0.046	0.117	−0.087	0.152
Intrastate	**−0.032**	**0.072**	−0.179	**0.099**
Owner-operator	−0.112	0.064	−0.002	0.082
Private fuel	**0.439**	**0.040**	−0.082	0.058
Share wholesale	**0.0052**	**0.0021**	0.0009	0.0026
Share retail	−0.0021	0.0029	0.0015	0.0039
Missing sales/inventory	**0.389**	**0.081**	**−0.459**	**0.114**
Missing share wholesale/retail	0.046	0.096	0.223	0.133
Omega1	−0.979	2.8E + 07		
−Log L		5732.76		

Difference denotes the difference in the parameter estimates on the EVMS and trip recorder branches. Boldface indicates rejection of H_0: $\beta = 0$ using a two-tailed t-test of size 0.05.

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the coefficient on refrigerated van remains positive and significant, there is less evidence from this subsample that trip recorders are more valuable when trucks deliver to loading docks than when they do not. The main conclusions from Table II carry over, however. Monitoring's incentive benefits are higher when trucks stop infrequently, when on-time arrivals are particularly important, and when trucks haul dangerous or hazardous cargo.

The estimates of $(\beta_3 - \beta_2)$ are qualitatively similar to those in Table II but provide weaker evidence regarding the hypothesis tests. The coefficient on tank truck remains negative, but is now statistically significant using a t-test of size 0.10, but not 0.05. Those on the private fleet and contract carriage dummies also remain negative, but the latter is now not statistically different from zero. The negative and significant estimate on the contract carriage dummy in Table II may reflect the effect of carriers' unobserved technological capabilities rather than differences in the value of monitoring. In general, however, the estimates here suggest the same qualitative conclusions regarding OBCs' incentive and coordination benefits as above: OBCs' benefits are disproportionately incentive-related when verification is important and coordination-related when dispatchers' resource allocation decisions are constrained.

VI. Conclusion

There are long-standing controversies about the organizational impact of IT adoption. Debates about whether IT leads to larger or smaller firms, more or less centralized decision-making, and more or less empowered workers continue to this day.[19] These debates reflect theoretical and empirical difficulties. The theoretical difficulty is that while existing theories sometimes provide sharp predictions regarding the organizational implications of improvements in particular informational capabilities, IT usually offers multiple new capabilities. As in the case of monitoring technologies, these capabilities often have opposing organizational implications. Debates arise because IT's net organizational effect is often theoretically ambiguous. The empirical difficulty is that analysis has been impeded until recently by the lack of detailed systematic data about firms' IT and organizational

19. These debates start with Leavitt and Whisler [1958]. See Malone, Yates, and Benjamin [1987], George and King [1991], and their references.

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characteristics.[20] The fact that organizational characteristics are often difficult to quantify compounds this problem.

Understanding the microeconomic implications of IT adoption requires empirical work on two complementary fronts. First, work is needed to test existing theoretical propositions. Proper tests generally should be based on individual informational capabilities, not IT's overall effects. Second, more work is needed to measure general relationships between IT and organizational change. Research on this second front generally does not test the validity of individual theories. Rather, it establishes the relative importance of different theories for understanding IT's overall effects. Further research on these fronts will make academic and public policy debates regarding IT's effect on the economy better-informed.

This paper contains work on the first front. I demonstrate the salience of agency theory regarding monitoring's incentive benefits, and show that contrasts between agency and decision theory contain insights for where monitoring's benefits are primarily incentive- or coordination-related. These results suggest where networking applications are likely to change workers' productivity, compensation, and job definitions. By showing where monitoring's incentive- and coordination-improving capabilities are important relative to each other, they also suggest how networking applications' impact on job definitions is likely to vary across different decision-making environments.

APPENDIX 1

This Appendix describes the process whereby I obtain the sales/inventory ratio and distribution of manufacturers' shipments series used in this paper. The main complication is that the product and business establishment categories used in other surveys do not correspond exactly to the product categories used in the TIUS. In several cases (for example, "mixed cargo"), it is impossible to link TIUS product codes to categories used in these other surveys.

Details regarding how the codes used in these other series were assigned to TIUS product codes are available from the

20. Along with this paper, recent work that has used systematic data includes Brynjolfsson and Hitt [1997] and Ichniowski, Shaw, and Prennushi [1997].

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author upon request. Appendix 2 reports the series used in the paper.

Sales / Inventory Ratios

The 1992 Census of Wholesale Trade Geographic Area Series contains sales and inventory data for wholesalers, by the type of commodity they generally handle (four-digit SIC codes 50xx and 51xx). The scope of the data includes wholesale establishments only; it does not include warehouses of manufacturers or retailers unless they are considered separate establishments.

The wholesale SIC codes are linked to TIUS product categories where possible. For each TIUS product category for which such a calculation is possible, I obtain figures for total 1992 sales by wholesalers and end-of-year 1992 inventories at wholesalers by aggregating across the SIC codes within that product category. I then obtain sales/inventory ratios for wholesalers handling products in each TIUS product category from these aggregates.

These calculations reveal that the sales/inventory ratio for wholesalers handling live animals was approximately 190: over five times that in any other product category. Handling and managing inventories of live animals at stockyards is fundamentally different than those handling other (nonliving) goods. I handle this outlier by dropping it from the series and treating the value as missing.

Distribution of Manufacturers' Shipments

The 1987 Census of Manufactures Subject Series contains data on manufacturers' distribution of sales by class of customer. No such data are available in the 1992 Census of Manufactures. These data are available for all two-digit SIC categories and many four-digit SIC categories. The data report the proportion of shipments, by value, which are to internal and external customers. They also report the proportion which are to wholesalers, retailers, manufacturers, and governments.

Series are obtained for TIUS product categories in a fashion similar to above. SIC codes are matched to TIUS product categories where possible. Proportions shipped to wholesalers and retailers by TIUS category are calculated after aggregating across the individual SIC categories that make up each TIUS product category.

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APPENDIX 2: VALUE, SALES/INVENTORY RATIOS, AND DISTRIBUTION OF MANUFACTURERS' SHIPMENTS

	Value/weight ($000/ton)		Sales/inventory ratio (Wholesalers)	Value share of manufacturers' shipments	
	Private	For-hire		To wholesalers	To retailers
Farm products	0.36	0.23	19.73	—	—
Live animals	1.30	1.44	—	—	—
Processed food	0.96	1.25	27.12	56.1%	23.8%
Mining products	0.07	0.08	24.85	—	—
Building materials	0.03	0.06	13.44	—	—
Logs and other forest products	0.03	0.11	26.53	—	—
Lumber and fabricated wood products	0.21	0.31	13.44	47.3%	13.8%
Paper products	0.68	0.86	20.09	19.8%	15.8%
Chemicals	0.50	1.25	17.50	29.6%	12.2%
Petroleum	0.22	0.18	32.41	75.7%	7.6%
Plastics and/or rubber	2.02	2.63	24.11	26.5%	11.5%
Primary metal products	0.61	0.77	—	19.7%	1.5%
Fabricated metal products	1.22	1.76	11.57	24.2%	5.1%
Machinery	3.85	7.25	5.98	28.3%	8.9%
Transportation equipment	2.97	4.42	17.50	12.5%	32.9%
Furniture or hardware	3.06	3.13	12.18	28.4%	51.6%
Textile mill products	5.32	8.67	10.60	18.1%	40.8%
Glass products	1.72	1.04	—	26.3%	7.7%
Miscellaneous products	4.44	7.25	—	27.9%	13.1%

See Appendix 1 for sources and calculation of series. All TIUS categories not listed (e.g., mixed cargo) have missing values for each series.

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